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Post-Miocene Clockwise Rotation of the Miura Peninsula and Its Adjacent Area

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The Izu Peninsula is situated at the northern edge of the Philippine Sea plate (SUGIMURA, 1972), and is believed to have collided with the Honshu Island in the Early Quaternary (MATSUDA and UYEDA, 1971; MATSUDA, 1978). If the collision had occurred, it should have given rise to the deformation of the areas around the Izu Peninsula. In order to examine how much the collision deformed the area to the east of the Izu Peninsula, paleomagnetic work was carried out on the Miocene strata distributed in the Miura and Oiso area (Fig. 1). We also performed an age determination study of the strata using the fission track method.

Geology of the studied area was summarized by MITSUNASHI *et al.* (1979). According to them, the Miura Peninsula and Oiso coastal area are comprised mostly of Miocene to Pliocene marine sediments which are mainly composed of tuffaceous siltstones and sandstones. The strata in the Miura Peninsula are stratigraphically divided into two groups, namely lower to middle Miocene Hayama Group and middle Miocene to Pliocene Miura Group. The Hayama Group occurs in the Hayama-Mineoka zone, which is situated in the middle part of the Miura Peninsula and extends to the Boso Peninsula. The Miura Group unconformably overlies the Hayama Group. The stratigraphy of the Miura Group has been established in detail from many interbedded marker tuffs. An alternation of siltstone and sandstone strata is exposed along the Oiso coast. The strata belongs to the Miura Group (KOZIMA, 1954). The age of the formation was assigned to the N 17 of Blow's planctonic foraminiferal zonation (IBARAKI, 1977), which corresponds to late Miocene.

Fission track dating was carried out on two horizons in the Miura Group; one (MI 20) is the Hk tuff (MITSUNASHI *et al.*, 1979) in the upper part of the group, and the other (MI 31) is a tuff layer in the middle part of the group. Sampling localities for fission track dating are indicated in Fig. 1. The method employed in this work is grain-by-grain re-etch subtraction method on zircon crystals (NISHIMURA, 1981). Mean ages and their standard errors (GREEN, 1981)

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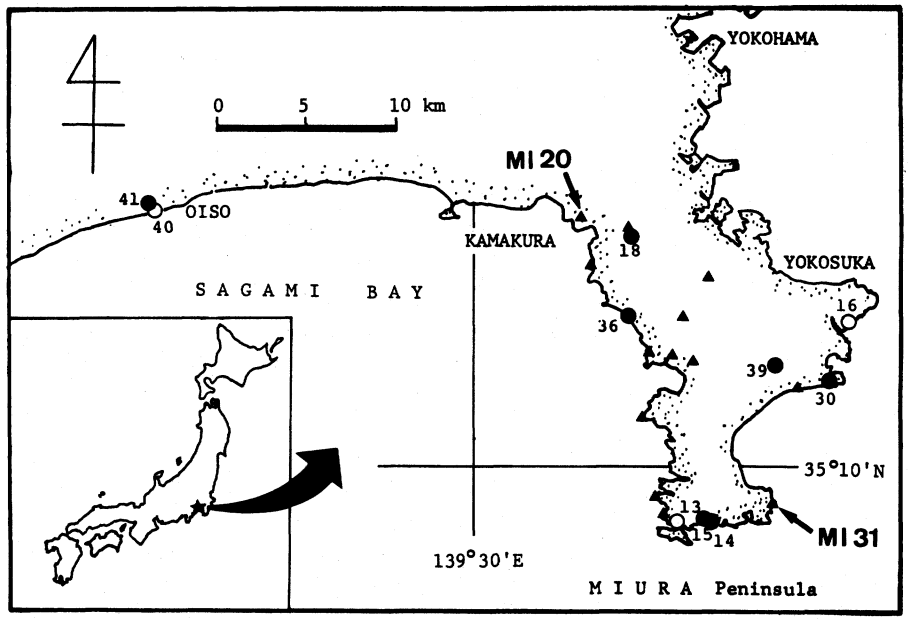


Fig. 1. Map showing the sampling localities. Circles and triangles are the localities giving magnetically stable and unstable samples, respectively. Open and solid circles show normal and reversed polarity, respectively. Arrows indicate localities of samples for fission track dating. Numerals denote locality numbers.

of the samples MI 20 and MI 31 are 3.7 ± 0.3 Ma and 5.1 ± 0.5 Ma, respectively. These ages are concordant with the micropaleontological information (TSUCHI *et al.*, 1981).

Samples for paleomagnetic work were collected from 55 sites on 21 localities in the Miura Peninsula and from 4 sites on 2 localities in the Oiso area (Fig. 1). The 55 sites in the Miura Peninsula cover from the Lower part of the Hayama Group to the upper part of the Miura Group. The rock types are tuff or tuffaceous siltstone. One to five oriented hand samples were collected from each site. Several cylindrical specimens of 1 inch diameter and length were prepared from each hand sample in the laboratory using a diamond core drill. Remanent magnetization was measured using a Schonstedt SSM-1A spinner magnetometer. Stability of natural remanent magnetization (NRM) was tested using alternating field (a.f.) and thermal demagnetization.

For each site, more than three specimens were demagnetized by a.f. progressively. Several additional specimens were demagnetized on the step giving the largest precision parameter in the set of progressive a.f. demagnetization. Data from 35 sites were abandoned by the following reasons: (1) Precision parameter was small (less than 10) throughout the demagnetization steps (3 sites). (2) The largest precision parameter was given at NRM step and was not im-

proved by the demagnetization treatment (4 sites). (3) The mean direction before tilting correction could not be distinguished from the direction of the present geomagnetic field (28 sites). (4) The site mean direction was evidently apart from the others (more than 45°) and thought to be transitional on the way of a magnetic reversal (3 sites). 24 sites including 4 reversely magnetized ones survived consequently.

Thermal demagnetization experiments were also performed on at least one specimen for each site. The remanent magnetization direction behavior for all the sites which passed the above criteria were similar in both a.f. and thermal treatments. Pilot specimens from two of the rejected sites gave stable end point on the thermal demagnetization procedure, and its direction behaviors were different from that of a.f. treatment. On one of these sites, several additional specimens were thermally demagnetized at the step giving the largest precision

Table 1. Mean directions of the sites giving stable magnetization.

Area	Group	Locality No.	Site No.	Treatment	N	D_m	I_m	α_{95}	k
Miura	Miura	16	128	AF 100 Oe	9	−137.1	−49.8	5.1	104.7
		18	140	TH 200°C	1	18.9	34.0	—	—
			138	TH 200°C	7	14.8	39.3	4.7	164.5
		30	183	AF 100 Oe	5	22.3	17.7	6.3	147.8
			182	AF 200 Oe	7	25.8	25.8	6.5	106.0
		36	208	AF 200 Oe	8	21.8	24.2	4.7	142.7
			209	AF 150 Oe	8	30.6	30.6	6.6	71.4
			211	AF 150 Oe	4	29.3	27.4	3.5	685.5
		13	110	AF 300 Oe	7	−126.7	−70.9	3.7	270.3
		15	124	AF 150 Oe	7	12.4	27.3	10.5	34.0
			125	AF 100 Oe	19	10.4	32.1	3.7	83.4
		14	123	AF 150 Oe	3	12.0	25.0	11.7	112.0
			122	AF 150 Oe	7	12.6	30.5	3.8	225.4
			121	AF 100 Oe	8	4.0	19.0	4.3	167.2
			120	AF 100 Oe	4	6.3	50.8	5.8	255.3
			119	AF 150 Oe	7	16.0	34.7	5.7	112.8
			118	AF 100 Oe	8	23.0	24.7	2.1	729.9
			117	AF 150 Oe	7	15.7	36.7	4.2	210.2
			116	AF 200 Oe	7	22.3	29.7	4.5	179.1
			115	AF 100 Oe	6	17.8	45.4	2.5	717.3
			114	AF 100 Oe	7	18.6	44.3	3.0	406.2
	Hayama	39	225	AF 100 Oe	6	40.3	39.0	8.1	68.7
Oiso	Miura	41	235	AF 200 Oe	4	13.7	43.8	8.3	122.9
			233	AF 150 Oe	10	8.4	53.3	3.4	200.8
		40	228	AF 150 Oe	10	−140.8	−61.5	12.9	14.9
			229	AF 300 Oe	12	−121.5	−71.7	7.1	38.2

The data are shown in descending order on the Miura area. N, D_m, I_m, α_{95} and k are the number of specimens, mean declination, mean inclination, half angle of confidence cone of 95%, and precision parameter, respectively. AF and TH indicates alternating field and thermal demagnetization, respectively.

parameter in the set of the progressive thermal demagnetization. On the other site, there remained no specimen, so that the stable end point was regarded as site representative direction.

The site mean remanent magnetization directions are summarized in Table 1 and illustrated in Fig. 2. The mean direction was $D=18.3°$, $I=33.7°$, $\alpha_{95}=4.6°$, for 22 normal sites, and $D=-133.4°$, $I=-63.7°$, $\alpha_{95}=12.4°$, for 4 reversed sites. These two mean directions were 145.3° apart from each other, and significantly different from the other's antipodal direction. This difference would be explained by the presence of a stable secondary magnetization whose direction is parallel to the present geomagnetic field. If there remains a stable secondary component, it would bias the magnetic direction of both polarity to the same direction. Such an influence of a secondary component tends to cancel one another out by taking the mean direction of both polarity on the same weight (McELHINNY, 1973). Therefore, we calculated, in this study, total mean direction as follows; firstly, the mean direction of each of normal and reversed groups was calculated separately, secondly, reversed mean direction was translated to normal direction and, finally, the mean of the two mean directions was calculated. The total mean declination and inclination are, hence, 28.1° and 49.5°, respectively.

Declinations for all the sites giving stable magnetization were deviated clockwise. Because the 26 sites cover the time range of at least a few million years, they are considered to be enough to average out geomagnetic secular variation. Southwest Japan have experienced no rotation after the large clockwise rotation which occurred during the middle Miocene (TORII, 1983). This evidence suggests a clockwise rotation, as large as 30°, of the studied area. The absence of obvious difference in the site-mean direction throughout the sequence suggests that the rotation occurred after the deposition of the whole Hayama and Miura

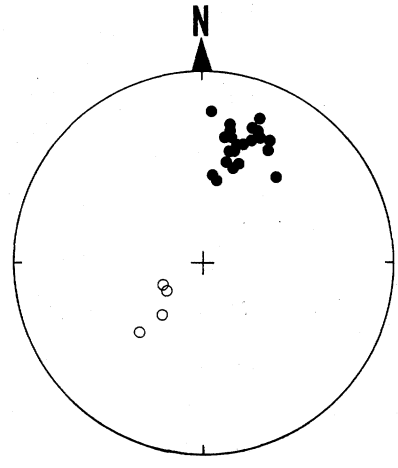


Fig. 2. Equal area projection of site mean directions. Open and solid circles denote upper and lower hemisphere projection, respectively.

Group. Therefore, it should postdate 4 Ma.

Based on geological evidence, MATSUDA (1978) proposed that the Izu Peninsula had collided with central Honshu in the early Quaternary and then moved northward for a few tens of kilometers. From the timing and the sense of the rotation of the studied area, it would be reasonable to consider that the rotation was associated with the collision of the Izu Peninsula. Although we have no direct evidence restricting the pivot position of the rotation, the pivot is considered to be located at the Uraga Strait (Fig. 3) for the following two reasons: firstly, the pivot should not be so far from the studied area as there is no evidence of a large northward drift of the studied area, and secondly the axis of Hayama-Mineoka Zone was deflected at the Uraga Strait. The northward component of the post-collision movement of the IZU Peninsula is, then, estimated to be about 30 km (Fig. 3).

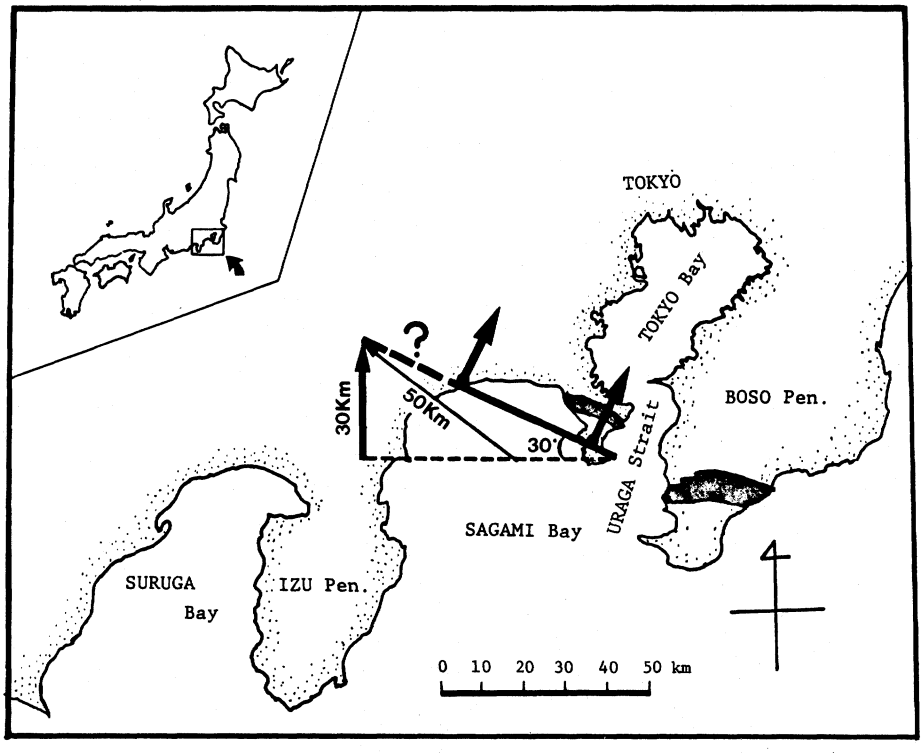


Fig. 3. Map explaining the relation between the rotation of the studied area and the collision of the Izu Peninsula. The rotation of the studied area around the Uraga Strait as large as 30° can be explained by about 30 km northward penetration of the Izu Peninsula. Assuming that the direction of the movement of the Izu Peninsula has been between north (MATSUDA, 1978) and N 55°W, which is that of the Philippine Plate (SENO, 1977), the post-collisional movement of the Izu Peninsula would be between 30–50 km. Shaded area shows the Hayama-Mineoka zone.

The direction of the movement of the Izu Peninsula is controversial. Since the Izu Peninsula is situated on the Philippine plate, it seems to move with the Philippine plate which moves toward N 55°W (SENO, 1977). On one hand, MATSUDA (1978) argued that the oblique subduction of the Pacific plate along the Izu-Bonin trench has dragged the Izu-Bonin arc to the north by incomplete decoupling between them. The post-collisional movement of the Izu Peninsula amounts to about 50 km in the former case, and 30 km in the later case. Assuming that the real movement had a direction between these two extremes, the Izu Peninsula moved 30–50 km after its collision. Since the collision occurred after 4 Ma, the velocity of the Izu Peninsula after the collision should have been more than 0.8 cm/yr, which is not inconsistent with the velocity of the Philippine plate (3 cm/yr; SENO, 1977).

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